

Military Asset Management System (MAMS)

Full-Stack Enterprise Defense Logistics & Inventory Tracking System Documentation

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Target Stack: Spring Boot 3 + React 18 + MySQL/H2	Security Standard: JWT + Zero-Trust RBAC

1. Project Overview

The **Military Asset Management System (MAMS)** is an enterprise-grade defense logistics platform designed to provide real-time visibility, inventory control, personnel assignments, inter-base asset dispatches, and unalterable security audit logging across strategic military installations. Built to ensure operational readiness, MAMS tracks every equipment movement—from initial procurement orders down to field expenditures.

Key Assumptions:

- Multi-Base Scoping:** Operational officers are assigned to specific military bases (e.g., Fort Liberty, Camp Pendleton), while Supreme Admins maintain command oversight across all bases.
- ACID Transaction Boundaries:** Stock decrements at source installations and stock increments at destination installations occur within atomic database transactions.
- Immutable Audit Log:** Transaction entries, personnel checkouts, and security logins publish append-only telemetry records.

Limitations & Operational Constraints:

- Telemetry Feed:** Real-time radar/GPS location feeds are simulated through REST API polling rather than direct satellite uplinks.
- Air-Gapped Deployment:** Offline field deployments require local H2/MySQL database synchronization upon re-establishing network connectivity.

2. Tech Stack & System Architecture

Layer	Technology / Framework	Selection Rationale & Capabilities
Backend	Java 21 LTS Spring Boot 3.2.5	High performance, strong static typing, enterprise ecosystem, native concurrency, and robust Spring Security integration.
Frontend	React 18 Vite 6 + Axios	Component-driven Single Page Application (SPA), lightning-fast hot module reloading, modular state context, responsive design.
Database	MySQL 8.0 / H2 In-Memory	Relational integrity with foreign key constraints, ACID compliance, zero-dependency embedded H2 fallback for instant cloud demo.

Security	Spring Security 6 JJWT (HS256)	Stateless JWT token authentication, granular RBAC method security (@PreAuthorize), BCrypt password hashing.
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Core Invariant Formulas:

Net Movement Formula:
Net Movement = Purchases + Transfers In - Transfers Out

Closing Balance Formula:
Closing Balance = Opening Balance + Net Movement - Expended Assets

Available Inventory:
Available Stock = max(0, Closing Balance - Assigned to Troops)

3. Core Data Models & Database Schema

Entity Table	Core Attributes	Relationships & Foreign Keys
military_bases	id (PK), code, name, location, commanding_officer, status	One-to-Many with users, inventories, purchases, transfers
users	id (PK), username, password, role, service_number, base_id	FK base_id -> military_bases(id) [Nullable for ADMIN]
assets	id (PK), asset_code, name, category_id, unit_price, is_expendable	FK category_id -> asset_categories(id)
base_inventories	id (PK), base_id, asset_id, opening_balance, closing_balance	Composite Unique (base_id, asset_id)
purchases	id (PK), purchase_order_number, base_id, asset_id, quantity, total_cost	FK base_id -> military_bases, asset_id -> assets
transfers	id (PK), transfer_number, source_base_id, destination_base_id, quantity	FK source_base_id & destination_base_id -> military_bases
asset_assignments	id (PK), assignment_code, base_id, asset_id, assigned_to_service_id	FK base_id -> military_bases, asset_id -> assets
asset_expenditures	id (PK), expenditure_code, base_id, asset_id, quantity, mission	FK base_id -> military_bases, asset_id -> assets
audit_logs	id (PK), timestamp, user_id, username, role, action, entity_name, details	Append-Only telemetry ledger for security & ops audit

4. Role-Based Access Control (RBAC)

MAMS enforces strict zero-trust Role-Based Access Control via **Spring Security filter chains** and **JWT claims**. Every incoming HTTP request is intercepted by JwtAuthFilter, which extracts the bearer token, validates the cryptographic signature, and sets the authenticated SecurityContext.

Role Name	Clearance Level	Permitted Operations & Scoping
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ROLE_ADMIN	Level 4 — Supreme HQ	Unrestricted CRUD access across all bases, user provisioning, global ledgers, and audit trail inspection.
ROLE_BASE_COMMANDER	Level 3 — Installation Command	Full operational oversight scoped strictly to their assigned military base. Can view audit logs for their base.
ROLE_LOGISTICS_OFFICER	Level 2 — Operational Logistics	Execute POs, dispatch inter-base transfers, check-out gear to troops, and log munitions expenditures.

5. API & Transaction Audit Logging

Every mutation state change (Purchase PO created, Inter-Base Transfer dispatched/received, Asset checked out/returned, Amunition expended, or Security Login) automatically dispatches an immutable audit event via AuditLogService. The log captures: **Timestamp**, **Actor User ID & Username**, **User Role**, **Action Category**, **Target Entity & ID**, **Detailed JSON Metadata**, and **Client IP Address**. The audit ledger is strictly append-only.

6. Setup & Installation Instructions

Local Development Quickstart:

1. **Prerequisites:** JDK 21+, Node.js 18+, Maven 3.9+ (or included mvnw wrapper).

2. **Backend Launch (Port 8081):**

```
cd backend ./mvnw spring-boot:run
```

3. **Frontend Launch (Port 5173):**

```
cd frontend npm install npm run dev
```

4. **Windows One-Click Launcher:** Simply run .\start.ps1 or .\start.bat from the root project directory.

Cloud Deployment (Render + Vercel):

- **Backend on Render:** Deploy as a Docker Web Service using the included Dockerfile. Set PORT=10000.
- **Frontend on Vercel:** Import repository with Root Directory frontend. Set VITE_API_BASE_URL=https://.onrender.com/api.

7. Key REST API Endpoints Reference

Method & Endpoint	Role Required	Description & Output
POST /api/auth/login	Public	Authenticates credentials and returns JWT bearer token + user details.
GET /api/health	Public	System status check. Returns operational health metrics.
GET /api/dashboard/metrics	Authenticated	Calculates Opening, Net Movement, Expended, Assigned, and Closing balances.

GET /api/dashboard/net-movement-details	Authenticated	Audit breakdown of underlying purchases, inbound, and outbound transfers.
POST /api/purchases	LOGISTICS / ADMIN	Records PO, increments base stock, and logs audit event.
POST /api/transfers	LOGISTICS / ADMIN	Validates source balance, dispatches transfer, increments destination stock.
POST /api/assignments	COMMANDER / LOGISTICS	Checks out tactical gear to troop service ID; updates current balance.
POST /api/expenditures	COMMANDER / LOGISTICS	Logs expended ammunition/fuel; decrements live closing balance.
GET /api/audit-logs	ADMIN / COMMANDER	Fetches immutable transaction and security audit trail records.

8. Default System Login Credentials

All default pre-configured test accounts use password: password123

Username	Role	Assigned Military Base	Commanding Officer
admin	ADMIN	Central HQ (All Bases)	Gen. Alexander Hayes (4-Star General)
commander_liberty	BASE_COMMANDER	Fort Liberty Command (Base 1)	Gen. Marcus Vance (Lt. General)
commander_pendleton	BASE_COMMANDER	Camp Pendleton Marine Base (Base 2)	Col. Sarah Jenkins (Colonel)
logistics_liberty	LOGISTICS_OFFICER	Fort Liberty Command (Base 1)	Maj. Robert Evans (Major)
logistics_pendleton	LOGISTICS_OFFICER	Camp Pendleton Marine Base (Base 2)	Capt. Emily Rodriguez (Captain)
commander_nellis	BASE_COMMANDER	Nellis Strategic Air Base (Base 3)	Maj. Gen. David Sterling (Major General)

End of Military Asset Management System (MAMS) Technical Documentation